

INSTITUTE OF ACTUARIES OF INDIA

Subject SA2 - Life Insurance

November 2024 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- a) The duties and obligations of Appointed Actuary are set out within Section 7 of Schedule 1 (Actuarial Functions) Part II (Appointed Actuary) of the Insurance Regulatory and Development Authority of India (Actuarial, Finance and Investment Functions of Insurers) Regulations, 2024.

General duties and obligations are provided in Paragraphs 1 – 11; while the additional duties of an Appointed Actuary are specifically provided in Paragraph 12.

The additional duties of an Appointed Actuary of an insurer carrying on life insurance business shall include:

- i. Certifying the actuarial report and abstract and other returns as required under section 13 of the Insurance Act, 1938 (“Act”); [marks may be awarded as long as candidate identifies signing of ARA and other returns as a duty and obligation – with or without the specific para reference of the Insurance Act]
- ii. Complying with the provisions of the section 112 of the Act in regard to recommendation of interim bonus or bonuses payable by life insurer to policyholders whose policies mature for payment by reason of death or otherwise during the inter-valuation period; [marks may be awarded as long as candidate bonus recommendation as a requirement with details – with or without the specific para reference of the Insurance Act]
- iii. Rendering actuarial advice in respect of expenses of management of the insurer;
- iv. Ensuring that the premium rates of the insurance products are fair;
- v. Certifying that the mathematical reserves have been determined in the manner prescribed in the extant provisions of Insurance Regulatory and Development Authority of India (Assets, Liabilities and Solvency Margin of Life Insurance Business) Regulations, 2016 and taking into account the Guidance Notes /Actuarial Practice Standard issued by the Institute of Actuaries of India and any directions given by the Authority;
- vi. Ensuring that the policyholders' reasonable expectations have been considered in the matter of valuation of liabilities and distribution of surplus to the participating policyholders who are entitled for a share of surplus;
- vii. Submitting the actuarial advice in the interests of the insurance industry and the policyholders; (
- viii. Coordinating the calculation of mathematical reserves;
- ix. Ensuring the appropriateness of the methodologies and underlying models used, as well as the assumptions made in the calculation of mathematical reserves;
- x. Assessing the sufficiency and quality of the data used in the calculation of mathematical reserves;
- xi. Informing the Board of insurer about the reliability and adequacy of mathematical reserves;

[Max 7]

- b) The note can cover the following aspects:

Background

I. Description of product

Individual non-participating savings contract would normally have the following salient features:

- Benefits would generally be oriented towards providing a savings element to policyholders along with protection by way of a minimum guaranteed death benefit.
- A guaranteed maturity benefit would normally be payable as a lump sum on maturity or through series of regular payments.
- Typically, it is expected that total survival benefits would be higher than premiums paid.
- Premium and benefits would be fixed, guaranteed and known in advance
- Premium payment term and policy term would normally be from 5 years to up to 20-25 years.

Proposal - invest upto 15% in equities under individual non participating savings business

(2)

II. Aspects and implications

(i) Regulatory compliance and limits

- As per regulations – minimum 65% needs to be invested in fixed income securities. Thus, 35% can still be invested in other assets including equities. Thus, the proposal meets regulations
- However, there will still be regulations related to
 - Unlisted/other than approved investment can be upto maximum 15%
 - Criteria for investment in a company such as minimum dividend declared for a period, etc. (min 10% dividend for at least 2 years of three consecutive years for listed equities)
 - maximum investment in one stock (10% in total along with other conditions) and maximum 15% in one group company
 - limits on investment in a particular sector such as maximum 35% in BFSI sector or 15% in other sectors taking all asset classes into consideration
 - limits and criteria for investment in promoter group company such as not more than 5% provided it is listed etc. and other conditions

(2)

(ii) ALM

- Characteristics of liability
 - Liability is for a fixed term and could be for long durations
 - Guaranteed in money terms in local currency
- Characteristics of equity
 - Returns are not guaranteed
 - Returns may be volatile particularly over short term, though there may be longer term (expectation of) higher overall returns than fixed interest assets.
 - There is a risk of loss of capital or invested value due to inherent risk in equity investments
 - Equity investments are of no specified duration or tenor; how long investments are held depend on portfolio purchase/sell decisions of the fund manager
- Nature, Duration and Currency implications with respect to ALM
 - There is a mismatch in ALM from nature as well as term of assets vs liabilities
 - If equities invested in INR and product is also in INR, there is match in terms of currency
 - If the horizon for equity investment is for long term and that product also has a long term liability, can be considered a match to weed out the short term volatility as it is expected to generate returns higher than risk-free over long term
 - However, since the nature of liability is guaranteed whereas the returns of equity including principal is non-guaranteed, there is mismatch in terms of nature of liability
 - There will also a steady flow of dividend yield (non-guaranteed) over and above any expected capital appreciation on stock
 - The ALM mismatch is, thus, also a function of quality of stocks selected and the horizon for stock to be held.

(4)

(iii)

1. Profits

- Equities are measured at fair value in the balance sheet. Any change in unrealized gains is kept under Fair Value Change Account (FVCA).
- Thus, any unrealized gains in equities do not impact the P&L unless realized. Only dividend incomes are recognized within the P&L. In case the equity is realized, the unrealized gains/losses flow in the revenue account (and subsequently, to the P&L statement)
- This leads to loss on interest income from Fixed income securities (if otherwise, the same monies invested in such fixed income assets as opposed to equities)
- To that extent, the profits might be depressed as unrealized gains do not flow and dividend yields are typically lower than corresponding fixed interest coupons. However, if equities are actively traded to generated realized gains, there is a potential benefit to profits by having higher expected earnings compared to debt
- This does makes the P&L quite volatile if gains are realized actively, and losses are carried forward.
- However, if equities are held for long term and expected to earn higher returns than other fixed income securities, it is expected that profits should otherwise increase.

(Max 2)

2. Reserves

- As per regulations, the valuation rate of interest should take into account the composition of assets supporting the liabilities as well as the risks associated with investment in regard to receipt of income on such investment or repayment of principal. Further, it cannot be higher than the yields on assets maintained for the purpose of such category of contacts. Thereafter a suitable MAD needs to be applied.
- The assumed yield on equities might either be higher than risk free rate to allow for risk adjusted yield given expectation of higher returns or lower than risk free rate to allow for excess volatility in income including principal repayment as well as expectation of lower return
- Thus, depending upon time horizon, past performance, general market performance, and future outlook the discount rate used in reserving may need to be adjusted to reflect the asset mix for equity
- This might also lead to separate asset liability mismatch reserve as per APS2

(Max 2)

3. Solvency

Solvency ratio is defined as the ratio of available capital (ASM) to required capital (RSM). Required capital is a function of reserves and sum as risk. There might be two fold impact due to equity exposure

- RSM may increase or decrease due to change in reserve.
- ASM may fall due to lower earning in profits as unrealized gains do not flow
- Further, there is asymmetry in calculation of ASM with regards to equity investment. Note that ASM falls when equity market value falls, but ASM do not increase when equity market value increase. The gain or loss is checked at an aggregate level of total FVCA. Thus, any gains in equity is capped to 0 in FVCA, whereas any losses in equity (i.e. if FVCA is negative), such losses need to be allowed for, for reduction of ASM
- This also leads to volatility in solvency ratio due to movements in equity values
- Therefore, 15% equity investments should be considered only subject to sufficient headroom available in the ASM by the shareholder to absorb any potential future volatility in case equity values were to go down (even, if for the short term).

(Max 2)

4. Value

- Value can be defined from both in-force profits as well as value of new business

- If assessed on IEV basis, the VNB margins might not be effected materially. Even the VIF under IEV will remain unaffected largely. However, the actual net worth will get affected due to earnings as well as relative market values of equity vis-à-vis fixed income securities.
- Over long term it is expected that equities should give higher return than fixed income securities and thus, might lead to higher net worth accretion.
- EV operating earnings may be higher in case the expectation of higher equity returns are considered when setting expected real-world return assumption used to derive the unwinding impact. This may however, lead to higher economic variance while performing analysis of change of EV in case short term equity returns are volatile.

(Max 2)

(8)

c) Risks

The main risks by the company under this contract are

- **Market returns risk**

- Given only FMC, this becomes only source of income for the company and hence fund performance becomes a key. Higher returns will fetch higher FMC but lower mortality charges and vice versa.
- Also, if fund choices are available, this could also become a source of risk as funds with lower (expected) returns (for example, liquid funds) may be chosen by customer that will result in lower FMCs. Consequently, fund-mix would become a driver of net income as well.

- **Persistency risk**

- This affects the interplay between FMC, mortality charge and surrender charge.
- Given charges are capped by regulations both in % terms and an absolute cap on with no surrender charge after first 4 years, higher surrenders, especially in early years means non recovery of acquisition costs.
- This also means low FMC as lesser monies to invest in if low persistency
- However, higher surrenders also mean lesser number of lives reaching till maturity to get loyalty addition.
- Thus, this beneficial from profits/margins standpoint
- However, given the surrender charges on ULIP is significantly low and capped, it is highly unlikely that higher surrenders loss will be compensated by low loyalty additions in case of higher surrenders in the initial 4 years.
- However, higher surrenders after first 4 years where there is no surrender charge means low FMC but also means low loyalty additions.
- Thus, it is interplay between quantum of loyalty additions (i.e. mortality charges collected) vis a vis surrender charge in initial years or FMC collected over the duration of contract.
- Since mortality increase with age, thus profitability will be affected by age; should typically reduce with age, hence, higher persistency for higher ages can be source of risk

- **Additional mortality and persistency risks arising due to loyalty addition benefit**

- In effect, the mortality cover provided is free if all the survivors stay till maturity i.e 100% persistency (except of course those who died)
- This means the pooling of risk did not happen
- However, given not all policies will stay till maturity, thus, there will be cross subsidy between persistent and non persistent policies
- The charge for mortality needs to be higher than actual experience to pay for loyalty additions

- This also means that profitability might be dependent with age. If persistency is same across all ages and/or there are material deviations from expected experience, then higher ages will have higher loyalty additions and thus, may have lower profitability
- **Expense risk**
 - Expense amount mismatch risk – ie. The charges are not enough to recover the expenses. This would typically happen in early surrenders
 - Expense timing mismatch - as company will have acquisition cost including commission but not corresponding charge to match the timing of expenses (such as no allocation charge), the design of the product itself lends itself to greater new business strain – that could be exacerbated if more costs are incurred earlier in the policy term .
 - Inflation of expenses vs increase in fund value to have higher FMC income – Again expenses will increase in line with inflation (such as say, salary costs or policy maintenance expenses), the fund is also expected to grow at least in line with inflation to match inflation of expenses
- **Mortality risk**
 - Higher than expected deaths leading to risk of charge not being sufficient to compensate higher death cost
 - Lower than expected mortality means higher people being alive and staying till maturity to receive loyalty additions
- **Mis-selling risk**
 - Risk of selling unit linked contracts without explaining the risks fully to customer.
 - Including say selling to high age customer that mortality charges will be refunded without explaining fully that fund will deplete much faster due to high mortality charges leading to lesser amount getting invested in markets.
 - Or even that benefits may not be tax free if premium amount is more than 2.5 lacs say
- **Business Mix risk**
 - given surrender charges capped and that mortality charges are refunded this can be of significant risk, such as average premium size higher than planned leading to cap in surrender charges in absolute terms but commission is paid as percentage of premium
 - lower premium size leading to lower FMC per policy due to lower fund size than planned
 - age mix risk – as highlighted earlier profitability will be dependent upon age and hence mix risk and
 - cover multiple- i.e. SA chosen on death. If it is higher or lower than planned
- **Operational risk**
 - Risk of system not set up properly to ensure charges and benefits are in line with product construct as per pricing/policy document
 - risk of data not captured properly
 - another risk here is to not have historical mortality charges deducted being captured correctly for each policy and stored properly as separate variable.
- **Other risks**
 - Legal risk – contract wordings not clear, ambiguous or incorrect leading to litigation or even litigation if mis-sold to customers
 - Mis-pricing risk – risk that pricing not correctly including sensitivities around all assumptions and taking various model combinations

- Unit pricing risk – risk of NAV not being calculated correctly
- Volume risk - Higher than planned volume may lead to high NB strain and/or higher capital for solvency whereas lower than planned volume may jeopardize business plan and may not be able to recover product development and system costs

(Max 16)

d) Reserving

The reserving for unit linked contracts shall be as per the ALSM May 2016 regulations as amended as per IRDAI (Actuarial, Finance and Investment Functions of Insurers) Regulations, 2024

- The determination of the amount of liability under each policy shall be based on prudent assumptions of all relevant parameters. The value of each such parameter shall be based on the insurer's expected experience and shall include an appropriate margin for adverse deviations (hereinafter referred to as MAD) that may result in an increase in the amount of mathematical reserves.
- The amount of mathematical reserve in respect of a policy shall not be less than surrender value of the policy.
- Reserves in respect of Unit linked business shall consist of two components, namely, unit reserves and non-unit fund reserves.
- Unit reserves shall be calculated in respect of the units allocated to the policies in force at the valuation date using unit values at the valuation date.
- non-unit fund reserves shall be determined using discounted cash flow method, which shall take into account of the following, namely:-
 - premiums, if any, payable in future;
 - death benefits, if any, provided by the general (non-unit) fund reserve (over and above the value of units);
 - management charges paid to the general(non-unit) fund;
 - guarantees, if any, relating to surrender values or minimum death and maturity benefits;
 - Fund growth rates and management charges. (The values of these parameters, along with
 - others, shall be determined in accordance with para 5);
 - Non-negative residual additions, if any,
 - **Any future negative cash flow shall be appropriately provided for by setting up reserves; and negative reserves, if any, shall be set to zero.**

The loyalty additions can be reserve separately using discounted cashflow approach and added back to non-unit reserve or it can be reserved together under non-unit reserve

The reserve will be calculated on a per policy basis. For each policy, the reserve for loyalty addition will take into account the following:

- Past mortality charges deducted till date- the data should capture the total mortality charges deducted till the valuation date. This needs to be updated on every valuation date (say monthly)
- Future expected mortality charges to be deducted- This would require assumption on future expectations such are
 - Unit growth date and thus, FMC deducted
 - Persistency assumptions – how many people surrendering/withdrawing/paid up etc.
 - Mortality assumptions

For each time period, Unit growth rate and FMC deducted will help project the fund value and thus the SAR and thus, the mortality charges deducted. This will then be multiplied by probability in-force (i.e.

net of mortality and persistency) to determine the future mortality charges expected to be received each time period

Where underlying cash-flows for loyalty addition and other benefits are taken together as part of non-unit reserve, then future FMC and future surrender charges collected shall also form part of income of charges as above.

Expected In-force policies (net of persistency and mortality) till maturity will be eligible for loyalty addition. This will be multiplied by total mortality charges collected till date, past charges as on valuation date plus future expected charges

Where underlying cash-flows for loyalty addition and other benefits are taken together as part of non-unit reserve, then future expenses including future death outgo shall also form part of outgo to above.

Reserve for loyalty addition shall be calculated using discounted cashflow method - i.e. discounted value of outgo – less discounted value of Income, discounted as on valuation date.

Direction of MAD

- The direction of MAD would be considered as follows
 - Rise or fall in interest rate whichever is adverse
 - Increase or decrease in mortality whichever is adverse
 - Up and down in persistency whichever adverse.
- The direction of MAD will have interplay between:
 - High/low persistency and high/low interest rates
- Higher unit growth rate means higher FMC but lower mortality charges and thus, lower loyalty additions versus lower growth rate, thus lower FMC, but higher mortality charges and thus, higher loyalty addition
- Similar, higher persistency would mean higher no. of contracts in-force, leading to higher FMC and higher mortality charges, but this also means higher loyalty additions to be paid as higher people will remain in-force till maturity.

(Max 11)

[50 Marks]

Solution 2:

- a) The components of movement of EV should be bifurcated into Free Surplus, Required Capital, Value of In-Force

The analysis of change in IEV shall have the following components, each bifurcated into free surplus required capital and value of in-force; the sum total of which is EV:

- i. Opening EV – value of EV at the start of period
- ii. Opening adjustments to EV, if any
- iii. **The sum total of above be called as adjusted opening EV**
- iv. Value added by new business during the period
- v. Expected return on existing business
- vi. Transfers from VIF and Required capital to Free Surplus
- vii. Variance in operating experience split by major components including
 - mortality / morbidity
 - policy persistency
 - expenses, etc.
- viii. Change in operating assumptions
- ix. Other operating variance

- x. **The sum total of items iv to ix is the EV operating earnings or EV operating variance**
- xi. Economic variances – example - Any change in interest rates
- xii. Other non-operating variances
- xiii. **The sum total of operating, economic and other non-operating variances is the total EV earnings/ or total EV variances (i.e. items x to xii)**
- xiv. Capital contributions / dividend payouts
- xv. Closing adjustments
- xvi. **Closing EV – this is the sum total of opening adjusted EV plus total EV earnings plus capital contribution/dividend payouts plus any closing adjustment (i.e. items iii+xiii+xiv+xv)**

The disclosure should include an explanation of each step of the analysis of movement.

A reconciliation should be done of the adjusted net worth to the balance sheet as per the financial statements

(Max 8)

- b) Key reasons for relatively lower value of new business in par products compared to that under non-par products could be as following:

Segregation of funds: a life insurer is required to maintain separately:

- A life fund for participating policyholders, and
- A life fund for non-participating policyholders.

Sharing of surplus (participation by policyholders): The maximum surplus that shareholders are entitled to is:

- one hundred percent, in case of a non-participating fund;
- one-ninth of the surplus allocated to policyholders in case of a life fund maintained for participating policy holders.

Timing of distribution:

- In the case of non-participating fund, surplus may be distributed to shareholders at the relevant quarter/year end valuation during the period it emerges;
- However, in the case of participating fund, surplus may be distributed to shareholders only upon bonus declarations – which may potentially lead to a difference in timing of surplus emergence in the fund and surplus distribution to the shareholders.

In order to increase the profits under par contracts, the surplus allocated to policyholders needs to be increased so that the corresponding 1/9th of the shareholder proportion also increases. This needs to be managed together with the Company's wider par-fund management policies. However, some management actions that may be considered – subject to TCF, PRE and other par management principles could be as follows:

- **Maximizing distribution of surplus as it emerges without delaying distribution via the creation of Fund for Future Appropriations (FFA)**

This means no smoothening reserve and no FFA.

However, it will have challenges including freedom of investment and volatility of regular bonuses including volatile terminal bonus

- **Increasing actuarial surplus**

This means paying higher surplus through bonuses and reducing guarantees. In an extreme scenario, this could imply maintaining only the minimum death, maturity or surrender benefit as

guaranteed benefits as required by regulations; and paying any incremental benefits through bonus. This could include, for example, capital return on maturity (return of premium) as guaranteed benefits and pay everything else through bonuses.

However, this could lead to discomfort for customers, if the guaranteed levels are too low; especially if the product is different to what is offered by others in the market

This might also lead to reduction in benefits to customers who do not stay till maturity.

e.g. the death benefit, surrender benefit, paid up benefit might be lower in early years due to bonuses declared only for first few years and guaranteed element being low.

- **Distributing surpluses early to take benefit of tax losses including carry forward tax losses, if any**

If the pricing takes advantage of any accumulated tax losses which will get subsidized to the extent to surplus distributed and they might not incur any tax on surplus.

- **Increase asset share to support higher bonus**

- Revisiting expenses charged and/or mortality charged (by working on expense allocation/expense efficiencies and better underwriting controls. However, any expense allocation might lead to expense charged to other lines of business or lower new business volume if stricter underwriting controls are implemented.
- removing any cost of guarantee deducted. However, if such charge is removed, then there is risk of shortfall in fund in case any negative returns are faced or bonuses cannot be reduced materially due to PRE
- adding lapse/surrender/paid up profit surplus if not already done. This would have otherwise supported proxy to FFA which means fund is extremely tightly managed now
- utilizing FFA of the existing fund to support higher bonuses in the new product. This can be a risky approach especially when cross-subsidy is to such an extent to paying to different generation of policyholders; including may requiring either a prior regulatory approval or regulatory intervention.

(10)

c) Problem statement – managing a run off fund with significant FFA

Objective and considerations

- FFA should be distributed fairly across all generations of the policyholders
- PRE and TCF should be maintained
- Ensuring no regulatory challenges
- No bad press/publicity/litigation
- Impact to shareholder in terms of profits/returns
- Capital/solvency impact

Distribute whole or part of FFA

- i. Any such distribution should be aligned to internal document such as PPFM or any approach document for management of pension business, if available.
- ii. In a run off portfolio, the expenses needs to be projected keeping in mind the reduction of policies and thus, fund not able to support expenses, effectively leading to very high per policy costs. In such a case,

- iii. Any FFA distribution should not jeopardize the security of remaining policyholders in terms of obligation to pay guaranteed benefits (base benefits + bonuses declared) and/or limitation to declared future bonuses in line with PRE created, if any.
- iv. How to be paid, whom to be paid to when to be paid needs to be considered carefully and attempt should be made to avoid the tontine effect.
- v. If paid one off – should it be only to remaining policyholders or to policyholders that have exited previously as well.
- vi. If one – off – then need to decide, whether cash bonus or addition to benefit to be paid on death, maturity and / or surrender.
- vii. If paid wholly, then need to keep in mind it becomes guaranteed, whether paid now or later.
- viii. Any FFA payment might impact solvency as FFA is used as available capital for solvency calculation purpose.
- ix. Similarly, it will also restrict investment freedom as the company will now have less or no FFA to invest in riskier assets.
- x. Any FFA payment might also impact future bonuses if PRE is created for the same in case future returns do not support bonuses
- xi. Also, any format of distribution might bring dissonance to customers, say for those who exited early (due to death surrender. Maturity) and if FFA is not received to them.
- xii. Similarly, there might be irate customers who can litigate the company if distribution is not perceived as fair
- xiii. To determine the right amount of FFA applicable to each policy, asset share can be best reference.
- xiv. If distributed later, an uplift to asset share can be done.
- xv. Whom to distribute FFA will also be a function of sources of creation of FFA; is it lapse profits or cap in expenses due to EOM or under distribution of surplus
- xvi. Any FFA distribution might also require regulatory consent with proper justification.
- xvii. Any such distribution would require to be placed in with-profits committee as well as approval from the Board
- xviii. However, any distribution of FFA will also lead to immediate return/profits to shareholder which was otherwise tied up
- xix. One tertiary could be a marketing opportunity: this may help with brand building for the company if customers are delighted with such distribution.
- xx. The company might want to distribute it early to take advantage of any accumulated tax losses which will get subsidized to the extent to surplus distributed and they might not incur any tax on surplus. However, pension surplus is tax free and hence, might not have any benefit due to this.
- xxi. This can also support in boosting new business, for with-profits life business if customers might expect such future windfall for with-profits life business
- xxii. However, need to have proper disclosures and awareness to ensure no expectation for such one-off for other funds, say with-profits life fund
- xxiii. Or even for pension line, in case company might want to plan for new business under pension segment anytime in future.
- xxiv. However, any such one-off distribution should be done with proper disclosures to ensure there is no expectation created that this might be repeated in future in pension fund.

(13)

d) Problem statement – merging with life with-profits fund

Additional implications

- i. Size of FFA of pension fund and FFA of life with-profits fund
- ii. Given run off and significant FFA, this will most likely lead to upliftment of benefits of life with-profits policyholders
- iii. Tax implications given no tax on surplus of pension fund whereas tax on life fund
- iv. Cross subsidy of FFA between two funds and how fair is such cross-subsidy

- v. Even cross subsidy of expenses between two funds
- vi. Will bonus rates be similar or different
- vii. Whether investment pattern similar or different
- viii. Even asset share calculation methodology may be different – say one allowing for lapse/profit surrenders and other not
- ix. If case no SAR in pension fund, should the mortality profits/losses of life fund be shared with pension fund.
- x. Cross-subsidy between different generation of policyholder, or by size or terms between life and pension fund
- xi. Similarly, level of guarantees involved in both funds
- xii. Any FFA distribution should not jeopardize the security of policyholders in terms of obligation to pay guaranteed benefits (base benefits + bonuses declared) and/or limitation to declared future bonuses in line with PRE created, if any.
- xiii. Prior to merging the funds, a regulatory approval may be needed that may entail further compliance requirements.
- xiv. Similarly, a customer consent may also be required to do the same.

(7)

- e) The various parameters to be looked at in a reinsurance arrangement for such a business includes
- Type of arrangement – surplus, quota share risk premium, original premium basis, XOL etc. including facultative or obligatory from any of the parties - this will again a function of what kind of reinsurance arrangement available, at what cost and whether allowed by regulations.
 - Retention level – this will depend upon the company risk appetite, any regulatory requirement, the benefit of solvency etc. As per regulations, pure term business need to have a retention of min 25% on an overall basis. Further, reinsurance should not lead to fronting of business
 - Reinsurance rates. This will become key parameter point as this will drive the price of the product and cost for reinsurance need to be allowed for. This will also affect the competitiveness of product.
 - Underwriting standards – this is potentially amongst the most important considerations as an extremely cumbersome underwriting requirements or very stringent underwriting conditions may make the product unmarketable; even if competitively priced. Further, this also has to be considered in line with market practice
 - Guarantee of RI arrangement. Whether RI rates are guaranteed for the entire tenure of existing business. If not, there is a risk to Insurer in case the reinsurer has right to revise the rates for business already written but insurer has written the business and guaranteed terms (here, a single premium)
 - Reinsurers' Credit rating and risk of default – this needs to be looked at to ensure there is a strong trust factor on reinsurer and to reduce the counterparty default risk. Further, the regulations prescribe a minimum rating for the reinsurer (such as BBB for CBR)
 - RI rates and underwriting go hand in hand. A relaxed underwriting may lead to increase in price and vice versa. RI rates is function of many other parameters including type of arrangement, retention level, guarantee of rates etc.
 - Right of rejection of claim. Given section 45 allows no rejection of claim by whatsoever reason (except misstatement of age) after 3 years of policy issuance, the reinsurer should also be bound to equivalent considerations . If there is a mismatch of the terms versus section 45, then insurer is at risk

Other points to be looked at

- Level of error and omissions allowed related to reporting delay of claims and/or premiums
- Any exclusion to death except, for example, suicide, in the first year
- Payment of RI premium and frequency of payment.
- Any administrative issues from system standpoint for implementation of RI arrangement

- Any tax challenges; i.e. tax on payment of RI premium from which input credits cannot be claimed
- Technical support provided by reinsurer.
- Auto underwriting limits and claims limits- ie u/w limits and claims approval limit upto which the insurer can directly u/w the case or take decision on claim without referring to reinsurer
- Maximum limit on SA – any maximum limit on SA on each life basis above which the reinsurer may not be obligatory to take business

(Max 12)

[50 Marks]
